

TABLE I. COMPARISON OF LIGHTING SYSTEMS IN THE FIRST LIGHT BLOCK

| Scenar<br>io | First Block |       | Threshold |       | Data<br>Comparison |               | Status<br>Alert |
|--------------|-------------|-------|-----------|-------|--------------------|---------------|-----------------|
|              | Lux         | Power | Lux       | Power | Lux<br>meter       | Watt<br>meter |                 |
| 1            | 6156        | 20    | 3078      | 10    | 6150               | 20            | Idle            |
| 2            | 4500        | 9     | 3078      | 10    | 4635               | 9             | Trigger         |
| 3            | 4500        | 9     | 3078      | 10    | 4635               | 9             | Trigger         |
| 4            | 20          | 1     | 3078      | 10    | 19                 | 0.5           | Trigger         |

TABLE II. COMPARISON OF LIGHTING SYSTEMS IN THE SECOND LIGHT BLOCK

| Scenar<br>io | First Block |       | Threshold |       | Data<br>Comparison |               | Status<br>Alert |
|--------------|-------------|-------|-----------|-------|--------------------|---------------|-----------------|
|              | Lux         | Power | Lux       | Power | Lux<br>meter       | Watt<br>meter |                 |
| 1            | 6058        | 20    | 3029      | 10    | 6240               | 20            | Idle            |
| 2            | 5265        | 9     | 3029      | 10    | 5260               | 9             | Trigger         |
| 3            | 5265        | 9     | 3029      | 10    | 5260               | 9             | Trigger         |
| 4            | 21          | 1     | 3029      | 10    | 21                 | 1             | Trigger         |

TABLE III. COMPARISON OF LIGHTING SYSTEMS IN ALL LIGHT BLOCKS

| Scenar<br>io | Total |       | Threshold |       | Data<br>Comparison |               | Status<br>Alert |
|--------------|-------|-------|-----------|-------|--------------------|---------------|-----------------|
|              | Lux   | Power | Lux       | Power | Lux<br>meter       | Watt<br>meter |                 |
| 1            | 10600 | 40    | 5300      | 24    | 10650              | 40            | Idle            |
| 2            | 8120  | 30    | 5300      | 24    | 8145               | 30            | Idle            |
|              | 8120  | 30    | 5300      | 32    | 8145               | 30            | Trigger         |
|              | 8120  | 30    | 7500      | 24    | 8145               | 30            | Idle            |
| 3            | 8120  | 30    | 7500      | 32    | 8145               | 30            | Trigger         |
|              | 8300  | 30    | 5300      | 24    | 8345               | 30            | Idle            |
|              | 8300  | 30    | 5300      | 32    | 8345               | 30            | Trigger         |
|              | 8300  | 30    | 7500      | 24    | 8345               | 30            | Idle            |
| 4            | 8300  | 30    | 7500      | 32    | 8345               | 30            | Trigger         |
| 5            | 5900  | 20    | 5300      | 24    | 5723               | 20            | Trigger         |
| 6            | 5680  | 20    | 5300      | 24    | 5600               | 20            | Trigger         |

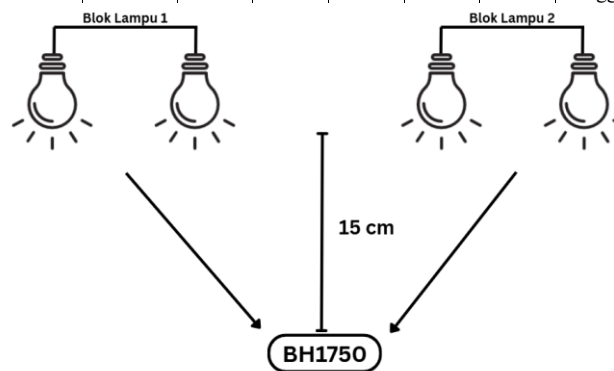


Fig. 3. Light Block Placement Scenario with BH1750 Sensor



Fig. 4. Blynk Notifications via Email

TABLE IV. AC TEMPERATURE CONTROL TESTING BASED ON SPECIFIED THRESHOLDS

| Initial Room Temperature DHT11 (°C) | High Temperature Threshold (°C) | Low Temperature Threshold (°C) | AC Temperature Setting (°C) | Final Room Temperature DHT11 (30-Minute Waiting Time) (°C) |
|-------------------------------------|---------------------------------|--------------------------------|-----------------------------|------------------------------------------------------------|
| 30.5                                | 27                              | 20                             | 24                          | 25.6                                                       |
| 27                                  | 27                              | 20                             | No Changes                  | No Changes                                                 |
| 27.1                                | 26                              | 20                             | 23                          | 24.3                                                       |
| 25.4                                | 26                              | 20                             | No Changes                  | No Changes                                                 |
| 23.7                                | 26                              | 20                             | No Changes                  | No Changes                                                 |
| 19.9                                | 26                              | 20                             | 23                          | 23.7                                                       |
| 23                                  | 29                              | 20                             | No Changes                  | No Changes                                                 |
| 21.2                                | 26                              | 20                             | No Changes                  | No Changes                                                 |

COMPARISON OF ROOM TEMPERATURE MEASURED BY DHT11 WITH TEMPERATURE MEASURED BY THERMOMETER AS REFERENCE MEASURING INSTRUMENT

| Initial Room Temperature Thermometer (°C) | Initial Room Temperature DHT11 (°C) | Final Room Temperature Thermometer (30-Minute Waiting Time) (°C) | Final Room Temperature DHT11 (30-Minute Waiting Time) (°C) |
|-------------------------------------------|-------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|
| 30.1                                      | 30.5                                | 25.3                                                             | 25.6                                                       |
| 26.6                                      | 27                                  | No Changes                                                       | No Changes                                                 |
| 26.7                                      | 27.1                                | 24                                                               | 24.3                                                       |
| 25.2                                      | 25.4                                | No Changes                                                       | No Changes                                                 |
| 23.6                                      | 23.7                                | No Changes                                                       | No Changes                                                 |
| 19.5                                      | 19.9                                | 23.3                                                             | 23.7                                                       |
| 22.8                                      | 23                                  | No Changes                                                       | No Changes                                                 |
| 21                                        | 21.2                                | No Changes                                                       | No Changes                                                 |

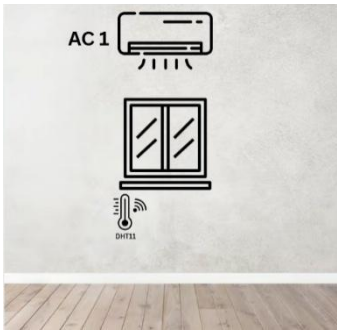


Fig. 5. DHT11 Placement Scenario

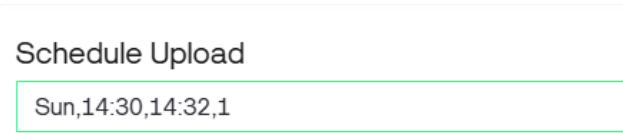


Fig. 6. Schedule Upload Feature in Blynk



Fig. 7. Device Status and Schedule Display when there is Schedule input

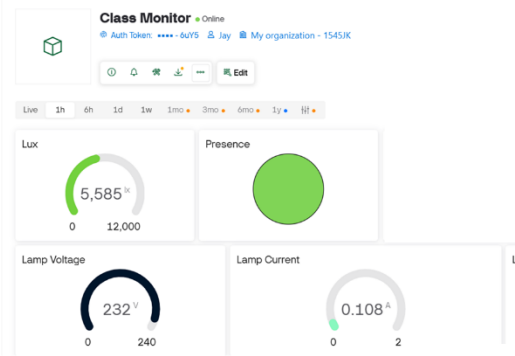


Fig. 8. Dashboard Blynk